

Oil Price Shocks, Central Bank Independence, and Inflation

Evidence from Cross-Country Panel Data

Pavan Sundar

ECC3479 - Research Project

May 2026

CBI stabilises Inflation Levels But Shock Filtering Is Untested.

Literature Review & Research Gap

- ▶ **Oil Shock Transmission:** Traditional macroeconomic frameworks argue that oil price increases raise domestic inflation through direct cost-push supply channels (*Blanchard & Galí, 2007*).
- ▶ **Institutional Credibility Framework:** Independent central banks can anchor inflation expectations by committing credibly to price stability and resisting politically driven monetary accommodation (*Cukierman, 1992; Romelli, 2022*).
- ▶ **Theoretical Mitigation Mechanism:**
 - ▶ Strong monetary institutions may prevent temporary commodity shocks from generating persistent wage-price spirals.
 - ▶ Independent central banks can signal a non-accommodative policy stance following external trade shocks.
- ▶ **Core Research Gap:** Existing literature explains either oil shock transmission or institutional credibility separately, but provides limited evidence on whether central bank independence actively dampens the short-run inflationary pass-through of global commodity shocks.

CBI may anchor expectations but will it filter external shocks?

Research Question & Hypotheses

Core Research Question:

To what extent do global oil price shocks affect domestic inflation, and does central bank independence mitigate their inflationary effects across countries?

- ▶ **Hypothesis 1 (Supply Shock Channel):** Positive global oil price shocks increase domestic consumer price inflation through the classical cost-push aggregate supply channel.
- ▶ **Hypothesis 2 (Institutional Mitigation Shield):** The inflationary impact of global oil price shocks is structurally smaller in countries characterised by higher degrees of central bank independence due to anchored domestic expectations.

Causal Ambition: Moving beyond correlations to identify whether structural monetary autonomy insulates open economies from external resource volatility.

Enforcing a 15-Year Contiguous Rule Optimises Panel Density.

Data Sources & Sample Attrition Pipeline

- ▶ **Longitudinal Baseline:** 1990–2023 panel timeline creating a filtered operational sample of **146 countries and 4,299 country-year observations**.
- ▶ **Data Quality Filter Pipeline:**
 - ▶ **15-Year Density Rule:** Restricts inclusion to cross-sectional units displaying ≥ 15 contiguous years, shielding panel parameters from short- T bias.
 - ▶ **Winsorization:** Capping inflation variables at the 1st/99th percentiles preserves the causal signal while neutralising outlying leverage.
- ▶ **Integrated Repositories:** Romelli (2022) CBI Dataset, IMF Inflation, and WB Pink Sheets.

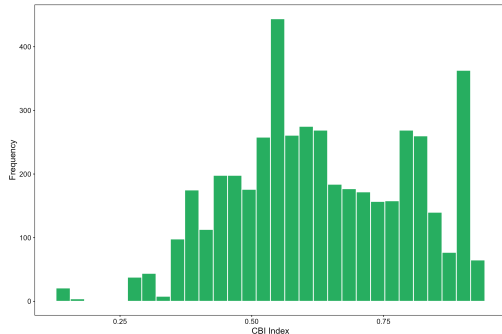


Figure: Sample Density: Variation Across the Legislative CBI Index

Lagged CBI and Fixed Effects Address Endogeneity Concerns.

Identification Strategy

Primary Interactive Two-Way Fixed Effects (TWFE) Layout:

$$\text{Inflation}_{i,t} = \beta_0 + \beta_1 \log(\text{OilP}_t) + \beta_2 \text{CBI}_{i,t-1} + \beta_3 (\log(\text{OilP}_t) \times \text{CBI}_{i,t-1}) + \gamma X_{i,t} + \alpha_i + \epsilon_{i,t}$$

- ▶ **Identifying Assumption 1 (Strict Exogeneity):** Global commodity benchmarks clear in international asset markets, operating exogenously to idiosyncratic domestic price innovations.
- ▶ **Volunteered Choice (Temporal Lagging):** Central bank reforms follow economic shocks. Applying a **one-year lag** ($\text{CBI}_{i,t-1}$) locks the institutional framework prior to current realisations, neutralising contemporaneous reverse causality.
- ▶ **Standard Error Structure:** Strictly clustered at the country level to correct for within-unit serial correlation and heteroskedasticity.

Oil Price Spikes Consistently Act as Global Demand Proxies.

Primary Regression Results Matrix

Dep Var: CPI	(1) Main Spec	(2) Baseline	(3) Country FE	(4) Crisis Robust
Log Oil Price (β_1)	-2.861*** (0.727)	-3.746*** (0.824)	-2.808*** (0.748)	-2.922*** (0.723)
CBI ($t - 1$, Std.) (β_2)	-5.241** (2.573)	-3.424 (3.340)	-5.699** (2.764)	-5.246** (2.576)
Interaction: Oil $\times$ CBI (β_3)	0.593 (0.640)	0.584 (0.778)	0.623 (0.677)	0.600 (0.641)
Interaction: Oil $\times$ Crisis				0.023* (0.012)
Observations	4,299	4,457	4,457	4,299
R-squared	0.384	0.051	0.345	0.384
Adjusted R-squared	0.362	0.050	0.322	0.362
Within R-squared	0.144		0.087	0.145
Lagged Inflation	Yes	No	No	Yes
Macro Controls	Yes	No	No	Yes
Crisis Controls	No	No	No	Yes
Country Fixed Effects	Yes	No	Yes	Yes

Our Core Baseline Signals Survive Extensive Robustness Stress-Testing.

Robustness Checks & Sensitivity Analysis

The null interaction effect ($\beta_3 \sim 0$) is structurally stable across all alternate measurements and functional checks.

It breaks **only** when isolating fixed exchange rate regimes, causing severe sample attrition and a structural break ($\beta_1 = 3.250^{***}$). Without nominal currency flexibility to absorb international trade benchmarks, global commodity volatility passes directly through to domestic consumer baskets via cost-push supply lines.

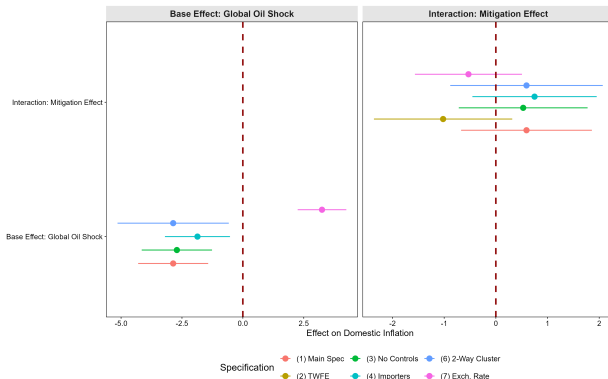


Figure: 95% Confidence Intervals Across Alternative Environments

CBI Anchors Nominal Targets But Fails to Cushion Real Shocks.

Conclusions & Policy Implications

- ▶ **Core Result:** Central bank independence is a powerful long-run disinflationary institution, reducing baseline inflation by approximately 5.24 percentage points ($\beta_2 = -5.241$).
- ▶ **Definitive Null Mitigation Effect:** The interaction between oil price shocks and institutional independence is statistically insignificant.
- ▶ **Implication:** Independent central banks can anchor expectations and stabilise long-run nominal conditions, but they do not function as short-run shock absorbers against global commodity volatility.
- ▶ **Key Structural Insight:** Once exchange-rate regime controls are introduced, the oil coefficient reverses into a large positive cost-push effect, demonstrating that external shock transmission depends primarily on currency frameworks rather than institutional autonomy.
- ▶ **Research Frontier:**
 - ▶ Decompose oil shocks into supply and demand components using **SVAR frameworks with sign restrictions** (*Kilian, 2009*).
 - ▶ Trace dynamic adjustment paths using **Panel Local Projections** (*Jordà, 2005*).

Thank You

Oil Price Shocks, Central Bank Independence, and Inflation

Questions & Comments

Pavan Sundar

Selected Academic References

Key Literature Citations

- ▶ **Blanchard, O., & Galí, J. (2007).** The Macroeconomic Effects of Oil Price Shocks: Real and Nominal Aspects. *National Bureau of Economic Research Working Paper No. 13368*.
- ▶ **Cukierman, A. (1992).** *Central Bank Strategy, Credibility, and Independence: Theory and Evidence*. MIT Press.
- ▶ **Kilian, L. (2009).** Not All Oil Price Shocks Are Alike: Disentangling Demand and Supply Shocks in the Crude Oil Market. *American Economic Review*, 99(3), 1053–1069.
- ▶ **Romelli, D. (2022).** The Political Economy of Reforms in Central Bank Design: Evidence from a new Dataset. *Journal of Economic Policy*, 3(112), 641-688.
- ▶ **Jordà, Ó. (2005)** Estimation and Inference of Impulse Responses by Local Projections. *American Economic Review*, 95(1), 161-182.

Acknowledgment on the use of Generative AI

In accordance with the Monash University Student Academic Integrity Procedure

I acknowledge the structural assistance provided by Generative Artificial Intelligence tools:

- ▶ **Tool System Utilised:** Google Gemini and Microsoft Copilot
- ▶ **Operational Scope:** Engaged for rendering mathematical LaTeX presentation layouts, validating balance sheet matrix metrics, and executing script checks across approximately 4 ideation iterations, 6 manuscript compression iterations, and 5 graphic layout debugging iterations.
- ▶ **Intellectual Verification Statement:** No programmatic outputs were blindly accepted. All empirical panel models, two-way fixed effects estimators, coefficient results, and econometric interpretations represent completely original work.